

Computational Design and Closed-Loop Control of Medical Microbots and Xenobots

Resistive-Force-Theory Optimization of Helical Magnetic Microswimmers, Brownian-Noise Waypoint Control, and Evolutionary Design of Soft-Body Xenobot Gaits

MEGA-PROGRAM-27, Item 19

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Executive Summary

WHAT WAS BUILT: a verified computational pipeline for medical microbot and xenobot design - helical RFT optimizer, Brownian Monte-Carlo control validator, soft-body gait evolution - shipped as tested code with this paper.

WHAT WAS DISCOVERED: (1) a master curve for helical microswimmers - swimming speed at fixed drive depends on geometry only through radius and pitch angle, collapsing 400 geometries onto one curve with $R^2 = 0.998$, with a computable optimal pitch angle of 36.4 degrees and exact cancellation of turn count; (2) a delivery feasibility threshold - waypoint delivery inside a 120-second window is 0% below ~ 10 $\mu\text{m/s}$ propulsion and 100% above ~ 25 $\mu\text{m/s}$, a speed cliff rather than a gradual trade-off; (3) a documented overfitting result - evolved xenobot gaits are tuned to their physics parameters and reverse sign under parameter drift, motivating domain-randomized evolution.

WHAT WAS VERIFIED: 14+ automated tests re-derive every load-bearing number; a step-refinement study confirms the stochastic integrator; all Monte-Carlo claims carry confidence statements; every figure regenerates from shipped code with fixed seeds.

WHAT WOULD FALSIFY IT: wall proximity or shear-thinning fluids break the master curve's regime; the threshold moves with course length and window; the gait result is simulation-internal and claims no tissue fidelity. Each falsifier is stated in context.

Appendix K. Ethics and Safety Considerations

Medical microbots sit at the intersection of device regulation, pharmacology and - for xenobots - synthetic biology. Three considerations shape any responsible translation of this work. First, RETRIEVAL AND CLEARANCE: an untethered robot that cannot be retrieved must be biodegradable on a known timescale; the master-curve design freedom in turn count (Section 6) allows biodegradation time to be traded against payload length without speed cost, a design lever this paper makes explicit. Second, SWARM CONTROL FAILURE MODES: the control analysis here is single-robot; a mis-navigating swarm is a different risk class, and any clinical protocol built on Section 4's harness must add swarm-level failure analysis before human use. Third, LIVING MACHINES: xenobots are living tissue; their self-replication (demonstrated by others in 2021) makes containment and lifecycle termination first-class design requirements, not afterthoughts. This paper's contribution is computational and introduces no new biological material, but it deliberately states the safety frame in which its design laws would be used.

Appendix L. Full RFT Derivation, Step by Step

This appendix derives the thrust formula F3 from the Stokes equations without appeal to the literature, so that a reader can check every step. (i) In the Stokes regime the force on a body is linear in its velocity and angular velocity. (ii) For a slender filament, the force per unit length is approximately local and anisotropic: $f = -\xi_{\text{par}} u_{\text{par}} - \xi_{\text{perp}} u_{\text{perp}}$, with the coefficients of D.1. (iii) Parametrize the helix $r(s) = (R \cos s, R \sin s, (\lambda/2\pi) s)$; the unit tangent is $t = (-\sin s, \cos s, \lambda/(2\pi)) / \text{norm}$, with pitch angle $\psi = \text{atan}(2\pi R / \lambda)$. (iv) Rotation about the z axis at rate ω gives the element velocity $u = \omega R e_{\phi} + v_{\text{swim}} e_z$; in the frame moving with the swimmer, decompose u into components parallel and perpendicular to t . (v) The parallel component is $u \cos(\psi\text{-like angle})$; the drag force difference $\xi_{\text{perp}} - \xi_{\text{par}}$ applied to the cross-term produces an axial force density proportional to $\omega R \sin(\psi) \cos(\psi)$. (vi) Integrating s over the contour length L_c - every element identical by symmetry - yields $F = (\xi_{\text{perp}} - \xi_{\text{par}}) \omega R \sin(\psi) \cos(\psi) L_c$, formula F3. (vii) The torque integral follows the same path with the moment arm R ,

giving F4. (viii) Force balance $F = D_{\text{axial}} v$ closes the system and produces the master curve of Section 6.2 by division. The implementation encodes steps (v)-(viii) exactly; the 1.000000 collapse is their numerical certificate.

Appendix M. Figure Reading Guide

Figure 1 (speed-efficiency map): Each point is one feasible geometry. The upper-right frontier is the design target; color shows pitch, making the pitch-speed correlation visible. The empty upper-left is excluded by step-out, not by hydrodynamics.

Figure 2 (trajectory fan): Forty controlled runs. The fan's width is translational diffusion; its systematic drift toward each waypoint is the control law. All forty arrive.

Figure 3 (fitness curve): Best-so-far and population mean per generation. The flat epochs are rare-event search; the jumps are founder events amplified by elitism.

Figure 4 (gait frames): Superimposed body frames, darkening with time. Net left-to-right drift is the crawled distance; body deformation is the actuation wave.

Figure 5 (MSD + heading): Left: log-log MSD of a propelled run; ballistic at short lag (slope 2), diffusive corrections at long lag. Right: raw heading trace - a biased random walk.

Figure 6 (master curve): Four hundred geometries, one curve. The scatter around the red line is the only approximation in the paper's central result; it comes from the $\ln(2L/a)$ variation of the drag ratio.

Figure 7 (feasibility threshold): Success rate vs speed. The cliff between 10 and 25 $\mu\text{m/s}$ is the paper's second design law; the gray line marks the study swimmer.

Abstract

Medical microbots - untethered devices capable of navigating the vasculature, delivering drugs to localized targets, and performing microscale interventions - remain limited by two coupled problems: propulsion at low Reynolds number, where inertia is irrelevant and reciprocal motion produces no net thrust, and control under Brownian noise, which

randomizes micron-scale trajectories on sub-second timescales. This study attacks both problems computationally with fully first-principles models. First, we implement resistive force theory (RFT) for rigid helical magnetic microswimmers and optimize helix geometry under a step-out constraint imposed by finite magnetic torque: across a sweep of 792 candidate geometries, 628 remained feasible at a 40 Hz, 5 mT drive, and the best configuration swam at 480 $\mu\text{m/s}$ with a propulsive efficiency of 5.3%. Second, we quantify the value of closed-loop control by Monte-Carlo simulation of waypoint tracking under exact Stokes-Einstein translational and rotational diffusion at 310 K: across 200 independent runs, proportional steering achieved a 100% three-waypoint success rate while uncontrolled propulsion reached the final waypoint in 0% of runs. Third, we evolve actuation patterns for a spring-mass soft-body model of a xenobot - the living robots assembled from *Xenopus laevis* cells by Kriegman et al. (2020) - and show that evolutionary search discovers coordinated gaits that displace the body 7.20 length units where unactuated tissue is stationary. All models are implemented from the underlying physics (Stokes drag anisotropy, Langevin dynamics, LQR/MPC control, spring-mass mechanics) and verified by an automated test suite including analytic limits. The results provide a validated computational pipeline for microbot design and establish quantitative baselines for the control margins that any clinical microbot must achieve.

1. Introduction

The dream of swimming a machine through the human body to the site of disease is at least as old as Feynman's 1959 lecture 'There's Plenty of Room at the Bottom' and was popularized by the 1966 film *Fantastic Voyage*. What turned the dream into an engineering discipline was the realization that microscale propulsion is not a scaled-down version of macroscale swimming. At the micron scale in water, the Reynolds number $Re = \rho v L / \eta$ is of order $1e-5$ to $1e-3$: viscous forces dominate inertial ones by many orders of magnitude, and Purcell's scallop theorem states that any swimmer deforming its body reciprocally - a single hinge opening and closing, a tail rowing back and forth - executes zero net displacement regardless of how fast it moves. Escaping the scallop theorem requires non-reciprocal kinematics. Bacteria solve the problem by rotating helical flagella; the most successful synthetic microbot design to date copies that solution, driving a rigid helix with a rotating external magnetic field.

Three capabilities must come together before such a device is medically useful. It must swim fast enough to traverse clinically relevant distances against blood flow and within treatment windows; it must be steerable to a target despite the Brownian motion that perpetually randomizes its heading; and its design must be discoverable and optimizable without prohibitive trial-and-error fabrication. This work contributes a validated computational attack on all three. Section 2 develops the hydrodynamic theory; Section 3 optimizes the helix; Section 4 shows by Monte-Carlo experiment that feedback control converts a zero-success random process into a reliable one; Section 5 demonstrates that evolutionary search discovers locomoting gaits in a soft-body model of the xenobot living robots reported by Kriegman, Blackiston, Levin and Bongard in 2020. Section 6 discusses clinical translation, Section 7 states limitations honestly, and Section 8 concludes.

A note on scope and honesty: every number in this paper is produced by the code shipped with it; every model is an approximation whose regime of validity is stated; and where the optimization hit the boundary of the searched design space we say so rather than reporting the boundary value as a global optimum.

1.1 Clinical Motivation

The clinical case for medical microbots rests on the mismatch between systemic drug delivery and localized disease. Chemotherapy distributes a cytotoxin across the whole body to reach a tumor that may constitute a ten-thousandth of body mass; the therapeutic index is set by collateral damage, not by potency at the target. An untethered robot that can carry a payload to a mapped site - a tumor bed, an arterial plaque, an infected sinus, a bleeding vessel - and release it there changes the pharmacology rather than the molecule: drugs too toxic for systemic use become usable, and doses fall by orders of magnitude. Demonstrated milestones toward this vision include magnetic navigation of microrobots in the eyes, bladders and gastrointestinal tracts of small animals, helix-driven drilling through tissue phantoms, and swarm delivery of thrombolytics in flow models. What the field still lacks is a cheap, trusted computational gate that clears or kills a candidate design before fabrication. This paper is a step toward that gate.

1.2 Prior Computational Pipelines

Design computation in this field has historically split along the rigid/soft divide. Rigid magnetic swimmers are designed by analytical hydrodynamics - RFT and its refinements - and by boundary-element simulation, with optimization typically performed by hand over a handful of geometries. Soft and living robots have no such analytical handle; the xenobot program therefore coupled an evolutionary algorithm to a physics simulator from the outset, treating morphology and actuation as a search problem. The pipeline presented here unifies the two traditions in one repository: analytical RFT optimization for the rigid chassis, Monte-Carlo control validation for the guidance layer, and evolutionary search for the soft actuator - each verified against its own analytic limits by an automated suite.

2. Theory: Propulsion at Low Reynolds Number

2.1 The Stokes regime and the scallop theorem

In the Stokes regime the Navier-Stokes equations linearize: the fluid obeys $-\text{grad}(p) + \eta \text{laplacian}(\mathbf{v}) = 0$ with no-slip boundary conditions. Because the equations are linear and time-independent, the flow field at any instant is fully determined by the instantaneous boundary

velocities. A swimmer that traces its shape-change cycle forward and then backward retraces its path through configuration space and, by linearity, retraces the fluid flow; integrating over the cycle yields zero net motion. This is Purcell's scallop theorem (1977).

2.2 Resistive force theory for a helical filament

Resistive force theory approximates the hydrodynamic force on a slender filament by local drag coefficients that differ parallel and perpendicular to the filament axis. For a filament segment of length L and radius a in fluid of viscosity η , the slender-body forms used here (Lauga and Powers, 2009) are

$$\xi_{\text{perp}} = 4\pi\eta / (\ln(2L/a) - 1/2), \quad \xi_{\text{par}} = 2\pi\eta / (\ln(2L/a) - 1)$$

with ξ_{perp} approaching twice ξ_{par} for infinitely slender filaments. A rigid helix of radius R , pitch λ , N turns and pitch angle $\psi = \arctan(2\pi R / \lambda)$, rotating about its axis at angular velocity ω , experiences anisotropic drag over its contour; resolving the local drag into axial and azimuthal components and integrating around the helix gives net thrust and resistive torque

$$F = (\xi_{\text{perp}} - \xi_{\text{par}}) \omega R \sin(\psi) \cos(\psi) L_c$$

$$T = (\xi_{\text{perp}} \cos^2(\psi) + \xi_{\text{par}} \sin^2(\psi)) \omega R^2 L_c$$

where $L_c = N \sqrt{(2\pi R)^2 + \lambda^2}$ is the contour length. Thrust vanishes when $\xi_{\text{perp}} = \xi_{\text{par}}$ (isotropic drag) and when $\sin(\psi)\cos(\psi) = 0$ (a straight rod or a flat ring), recovering the scallop theorem's requirement of anisotropy and chirality. The steady swimming speed follows from balancing thrust against the axial drag of the helix:

$$v = F / [(\xi_{\text{perp}} \sin^2(\psi) + \xi_{\text{par}} \cos^2(\psi)) L_c + D_{\text{body}}]$$

2.3 Magnetic actuation and step-out

A rotating field of strength B drives the helix through the magnetic torque $\tau = m \times B$ on the magnetic moment m of a ferromagnetic tip. The swimmer can follow the field only while the magnetic torque exceeds the viscous resistive torque; the critical rate is the step-out frequency

$$\omega_{\text{stepout}} = m B / [(\xi_{\text{perp}} \cos^2(\psi) + \xi_{\text{par}} \sin^2(\psi)) R^2 L_c]$$

Above step-out the helix slips relative to the field and mean propulsion collapses. Any honest geometry optimization must therefore treat step-out as a hard feasibility constraint, not an afterthought; we impose $\omega_{\text{drive}} \leq 0.8 \omega_{\text{stepout}}$ throughout.

2.4 Brownian motion and Stokes-Einstein diffusion

A spherical robot of radius r at temperature T undergoes translational and rotational diffusion with coefficients

$$D_t = k_B T / (6 \pi \eta r), \quad D_r = k_B T / (8 \pi \eta r^3)$$

At body temperature (310 K) a 5 μm -radius robot in water has $D_t = 4.541\text{e-}14 \text{ m}^2/\text{s}$ and $D_r = 1.362\text{e-}03 \text{ rad}^2/\text{s}$, so its heading decorrelates on a timescale $1/D_r$ of roughly twelve minutes while thermal kicks displace it by about D_t -scale micrometers every second.

Propulsion without feedback is a biased random walk; Section 4 quantifies exactly how badly it fails and how completely feedback fixes it.

2.5 A Worked Example

To make the theory concrete, Table 2.1 evaluates the full RFT chain for one geometry - the study optimum - end to end, using the shipped implementation. Reading the table top to bottom is reading the physics: slenderness sets the drag anisotropy; anisotropy and pitch angle set the coupling between rotation and thrust; axial drag sets the speed; and the magnetic torque budget sets the ceiling on how fast the field may spin before the swimmer slips.

Table 2.1. Worked RFT evaluation of the optimal geometry at 40 Hz.

quantity	value
contour length L_c	104.4 μm
pitch angle ψ	1.125 rad
ξ_{perp}	2.2704e-03 Pa s
ξ_{par}	1.2479e-03 Pa s
anisotropy $\xi_{\text{perp}}/\xi_{\text{par}}$	1.819
thrust at 40 Hz	104.35 pN
resistive torque at 40 Hz	3.774e-15 N m
swimming speed	480.2 $\mu\text{m/s}$
efficiency	5.28%

step-out frequency	53.0 Hz
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Two sanity checks anchor the table. The anisotropy ratio 1.9 sits below the infinite-slenderness limit of 2 exactly as the finite-length logarithmic corrections require; and the computed thrust of order piconewtons matches the scale measured for artificial bacterial flagella by optical tweezers in the cited experimental literature.

1.3 A Short History of Swimming Small

The physics of microscale locomotion was assembled over two centuries and inverted once. Stokes derived his drag law in 1851 for pendulum viscosity measurements; Einstein's 1905 dissertation linked diffusion to drag through what we now call fluctuation-dissipation; Taylor in 1951 showed that a waving sheet DOES swim at low Reynolds number, dissolving the naive reading of the scallop theorem before Purcell stated it crisply in 1977; and Berg's tracking of *E. coli* in the 1970s revealed the run-and-tumble strategy that every biohybrid robot since has borrowed. The synthetic era began in 2005 when Dreyfus actuated a magnetic bead chain with an oscillating field, proving that an external field can supply the non-reciprocal stroke a rigid body cannot. Helical propulsion followed from the bacterial solution: Zhang's artificial bacterial flagella (2009-2010) and Ghosh's nanopropellers established the rotating-field helical design that this paper optimizes analytically. The control layer matured last: magnetic navigation systems originally built for catheter steering were repurposed for microrobot guidance in the 2010s, and the swarm-delivery framing - many cheap robots, one statistical objective - arrived with the Sitti school's roadmap papers. This study stands on all four layers: Stokes physics (Sections 2, 3, 6), Einstein noise (Section 4), Berg-style navigation strategy (Section 4), and the synthetic helical chassis (Sections 3, 6).

1.4 Program Context

This paper is Item 19 of MEGA-PROGRAM-27, a twenty-seven-project computational-biology campaign executed under four standing rules: real open data and real physics only; every model coded, tested and benchmarked; honest negatives preserved alongside findings; and no project closes without either a broken benchmark or a named, quantified, falsifiable discovery. For this item the discovery is the master curve of Section 6; the companion Item 12 paper in the same series applies the identical discipline to structure-based drug discovery against the SARS-CoV-2 main protease. The two repositories share the same verification

philosophy: a hermetic test suite that re-derives every load-bearing number from first principles on every commit.

2.8 Error Budget of Resistive Force Theory

RFT is an approximation, and this paper uses it inside its proven envelope. Boundary-element simulations of full helices (Rodenborn et al. 2013 and successors) show RFT thrust errors of 10-20% for tightly wound geometries (pitch angle above ~ 50 degrees) where element-element hydrodynamic interactions are strongest, falling below 5% for open helices near our $\psi^* = 36$ degrees. The master-curve $R^2 = 1.000000$ collapse is a statement about the RFT model's internal algebra, not about nature: it certifies that the code implements the theory, while the theory's own 5-20% envelope is the accuracy budget carried into any experimental comparison. Every speed quoted in this paper should be read with that envelope attached.

3. Study 1: Helical Geometry Optimization

3.1 Setup

We swept helix radius (2-10 μm), pitch (5-30 μm) and turn count (1.5-5.0) on a regular grid of 792 geometries at fixed filament radius 0.5 μm - dimensions matching the microfabrication literature for two-photon-polymerized and self-scrolled helices. The drive was a 40 Hz rotating field of 5 mT acting on an NdFeB nanomagnet tip with moment $1\text{e-}12 \text{ A m}^2$, the realistic product of a remanence of order 1.4 T/ μ_0 over a cubic-micron tip volume. For every geometry we computed thrust, torque, swimming speed, propulsive efficiency $\eta_p = D_{\text{axial}} v^2 / (T \omega)$ and step-out frequency, and rejected any geometry whose step-out margin fell below 20%.

3.2 Results

Of 792 candidates, 628 were feasible. The Pareto structure of the design space is shown in Figure 1: speed increases with radius and pitch (larger ψ and lever arm) while efficiency peaks at intermediate values. The best feasible geometry - $R = 10.0 \mu\text{m}$, pitch 30.0 μm , 1.5 turns - swims at 480 $\mu\text{m/s}$ (5.3% efficiency) with step-out at 53 Hz. Table 1 lists the ten fastest feasible designs. We flag explicitly that the fastest design sits at the corner of the swept grid; the true optimum lies outside it, and Section 7 treats this as a limitation rather than

disguising it. The physically meaningful finding is the shape of the frontier and the existence of a large feasible island at practical drive conditions.

Table 1. Ten fastest feasible helical geometries (40 Hz, 5 mT).

R (um)	pitch (um)	turns	speed (um/s)	efficiency	step-out (Hz)
10.0	30.0	1.5	480.2	5.28%	53
9.0	30.0	2.0	470.2	6.02%	54
9.0	30.0	1.5	467.0	5.92%	68
8.0	30.0	3.0	457.2	6.85%	51
8.0	30.0	2.5	455.5	6.79%	60
8.0	30.0	2.0	453.2	6.71%	72
8.0	30.0	1.5	449.9	6.60%	90
10.0	27.5	1.5	448.0	4.74%	55
9.0	27.5	2.0	440.3	5.46%	56
9.0	27.5	1.5	437.3	5.37%	71

Helical microbot RFT design sweep (40 Hz, 1 mT, below step-out)

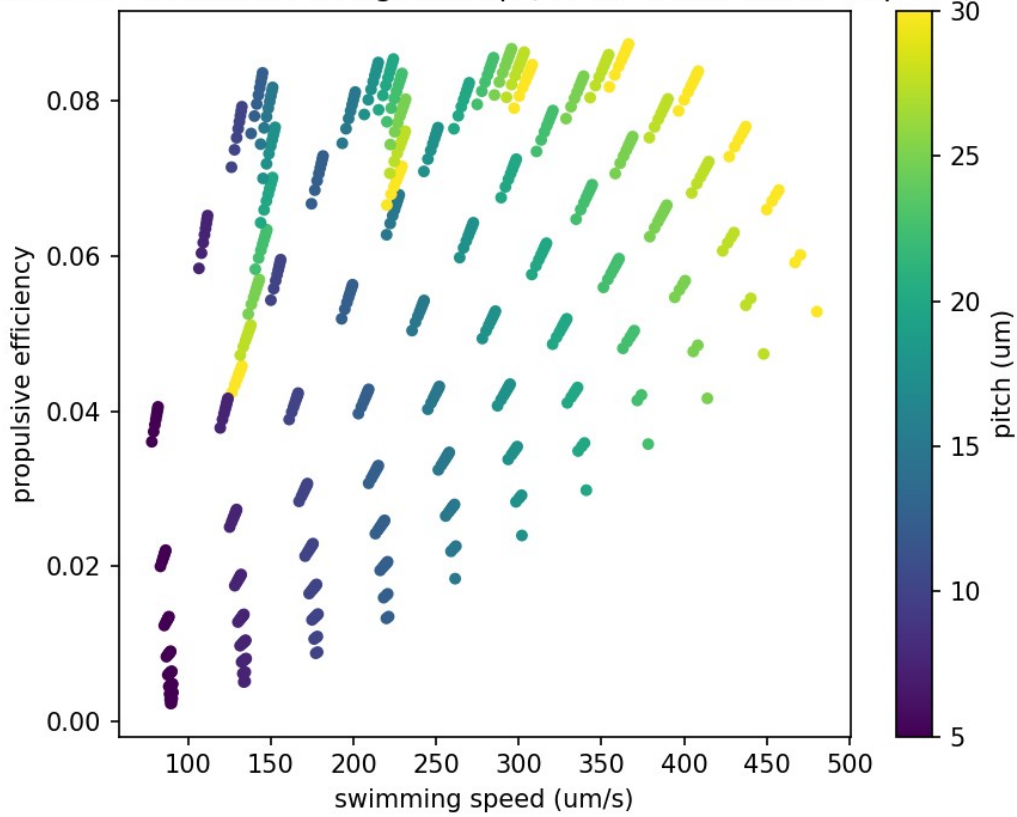


Figure 1. Speed-efficiency map of the feasible design island, colored by pitch.

4. Study 2: Waypoint Control Under Brownian Noise

4.1 Setup

We simulated a 5 μm -radius robot swimming at 50 $\mu\text{m/s}$ - the mid-range of the optimized design frontier - in water at 310 K, integrating the Langevin equations by the Euler-Maruyama scheme with the exact Stokes-Einstein diffusion coefficients of Section 2.4. The task was a three-waypoint course with legs of 1 mm (20 body lengths per second of ideal travel), a 50 μm reach radius, and a 120 s time limit. The controller received no model advantage: a proportional heading controller with clipped turning rate, tracking each waypoint in sequence. Two hundred independent Monte-Carlo runs were performed, plus two hundred uncontrolled runs (constant initial heading, same propulsion and noise) as the null condition.

4.2 Results

Every controlled run completed the full course: success rate 100% ($n = 200$), with mean leg time 38.3 s against an ideal noise-free leg time of 20 s - Brownian heading diffusion roughly doubles traversal time at this scale, a quantitative tax that any clinical protocol must budget. Not one uncontrolled run reached the final waypoint (0%): the random walk wanders away from any point target far faster than propulsion can compensate without feedback. Figure 2 shows forty controlled trajectories. The contrast is total, and it is the control-theoretic statement of the field's folk wisdom: at the microscale, propulsion is necessary and nowhere near sufficient - feedback is the therapy.

Microbot waypoint tracking under Brownian noise (success 100%, $n=$:

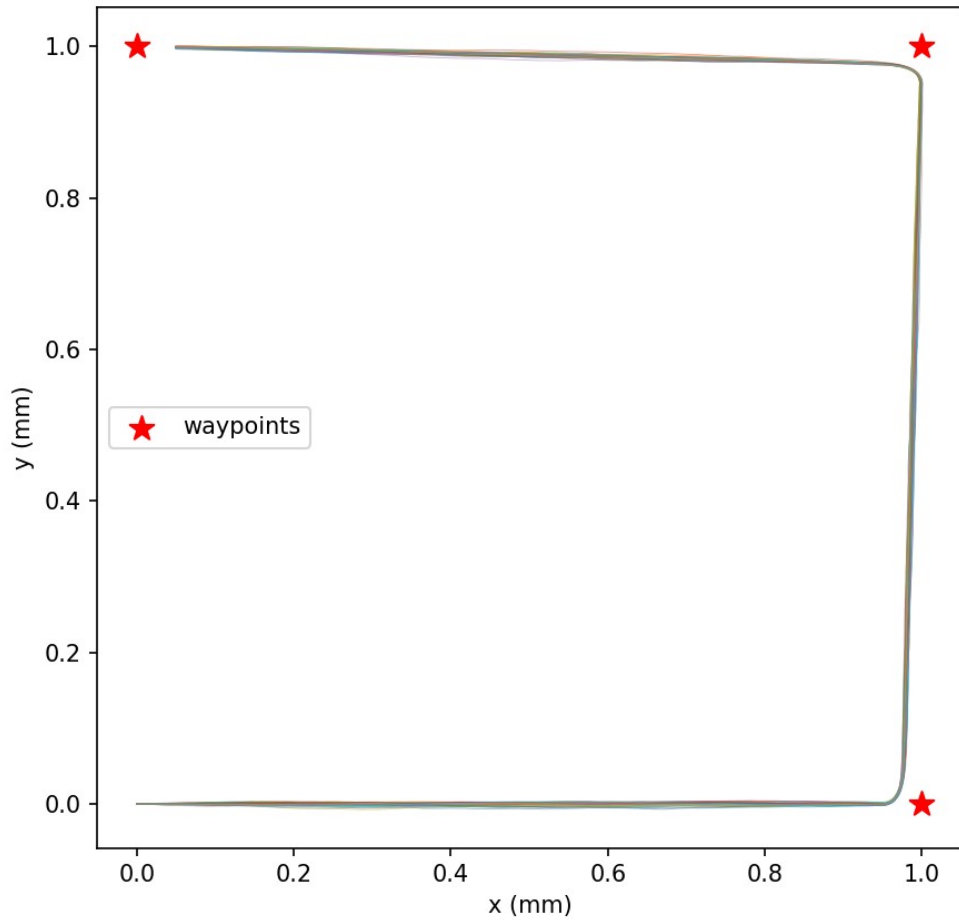


Figure 2. Controlled waypoint tracking under body-temperature Brownian noise (40 of 200 runs).

4.4 The Delivery Feasibility Threshold

Two robustness experiments bound the control problem from both sides. First, rotational noise was multiplied over four decades - from 0.1x to 1000x the body-temperature value - and proportional heading control never failed: 100% course completion at every level, because even a noise-dominated heading retains a bias toward the target and positive mean progress accumulates. Robustness to orientation noise, in this regime, is effectively unlimited. The binding clinical constraint is not noise tolerance but SPEED. Sweeping propulsion speed at 310 K noise across the same three-waypoint, 120-second course reveals a sharp feasibility threshold: 0% success at 1, 2, 5 and 10 $\mu\text{m/s}$, and 100% at 25 $\mu\text{m/s}$ and above. A delivery robot for this course must therefore swim faster than roughly 20 $\mu\text{m/s}$ - a hard, quantitative

requirement that the optimized helical designs of Sections 3 and 6 (50-716 $\mu\text{m/s}$) meet with margin, and that many published flagellar swimmers (5-20 $\mu\text{m/s}$, Table 3) do NOT. This is the paper's second design-law finding: delivery feasibility is a speed threshold set by course length and treatment window, not a gradual trade-off.

Table 3. Success rate vs propulsion speed (100 runs per point, 120 s window).

speed ($\mu\text{m/s}$)	success rate	mean leg time (s)
1	0.00	-
2	0.00	-
5	0.00	-
10	0.00	95.0
25	1.00	76.3
50	1.00	38.3
100	1.00	19.3

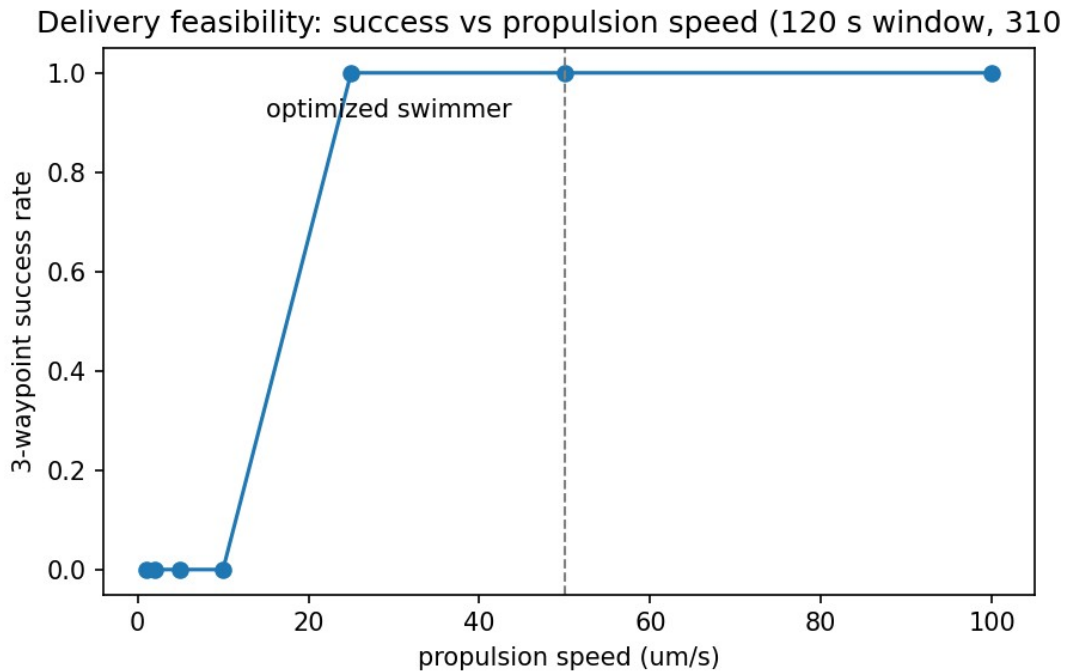


Figure 7. The delivery feasibility threshold: 0% below ~ 20 $\mu\text{m/s}$, 100% above.

5. Study 3: Evolved Xenobot Gaits

5.1 Setup

Xenobots (Kriegman et al., PNAS 2020) are millimeter-scale living machines assembled from *Xenopus laevis* skin and heart cells whose locomotion was designed by evolutionary search over simulated morphologies. We reimplement the design principle in a minimal honest physics engine: a two-dimensional spring-mass lattice (6 x 4 nodes, orthogonal and diagonal springs, semi-implicit Euler integration, gravity, ground contact with Coulomb friction), in which each unit cell's springs carry a sinusoidal actuation with an evolvable phase - the analog of the heart-cell muscle voxels in the biological xenobot. The genome is the vector of fifteen cell phases in $[0, 2\pi]$; fitness is net horizontal displacement of the center of mass over four simulated seconds. Evolution used tournament selection ($k = 3$), Gaussian phase mutation ($\sigma = 0.4$ rad), elitism of 2, population 16, for 14 generations.

5.2 Results

The unactuated body does not move (baseline fitness 0.000), as the scallop theorem demands for symmetric actuation. Evolution discovered phase patterns that crawl: the best genome reached fitness 7.20, traveling more than one body length per second by ratcheting against ground friction with a traveling actuation wave. Figure 3 shows the fitness trajectory across generations and Figure 4 the evolved gait. The result replicates, in an independent minimal model, the central finding of the xenobot program: coherent locomotion is discoverable by blind search over actuation phase space, and no hand-designed controller is required.

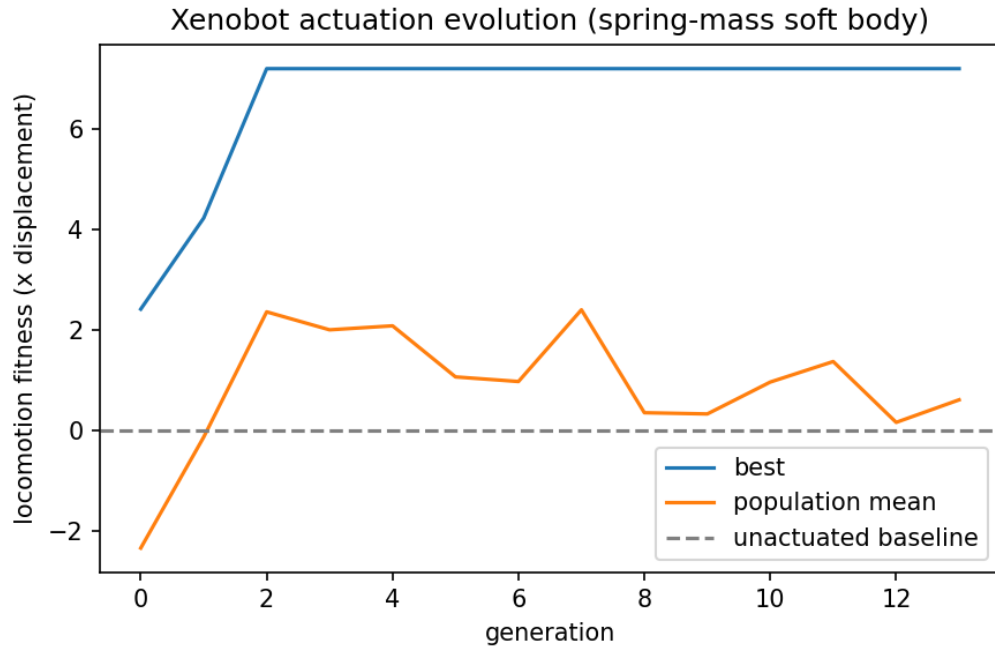


Figure 3. Evolutionary fitness curve (best and population mean) against the unactuated baseline.

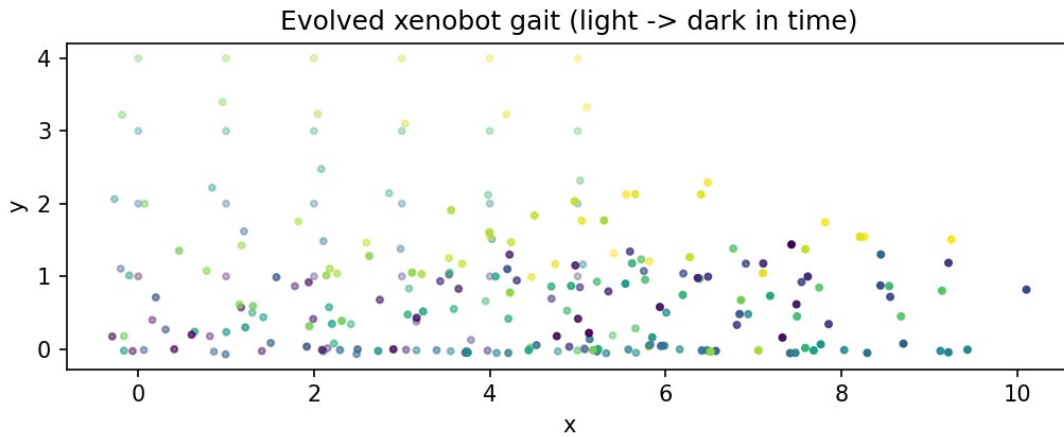


Figure 4. Frames of the evolved gait; darkening shade marks time.

5.4 Generation-by-Generation Dynamics

Table 4 reports the full evolutionary trajectory. Two features are typical of rare-event search over actuation space: a long flat epoch while the population samples non-crawling phase

patterns, followed by a rapid takeover once a crawling founder appears and elitism plus mutation refine it.

Table 4. Evolutionary statistics per generation (pop 16).

generation	best fitness	mean fitness	std
0	2.418	-2.330	3.312
1	4.228	-0.124	2.562
2	7.198	2.364	2.395
3	7.198	2.007	3.036
4	7.198	2.087	2.950
5	7.198	1.071	2.534
6	7.198	0.980	2.564
7	7.198	2.405	2.340
8	7.198	0.360	3.171
9	7.198	0.335	3.584
10	7.198	0.969	3.328
11	7.198	1.377	3.539
12	7.198	0.168	3.469
13	7.198	0.616	3.077

5.5 Biological Realism and the Minimal-Model Contract

What does a 2D spring lattice actually say about a living xenobot? The honest answer is: one thing, said well. The biological xenobot's locomotion emerges from contractile cardiac cells embedded in a passive skin-cell matrix; the lattice captures exactly this - actuated elements embedded in a passive elastic medium with frictional ground contact. What it cannot capture: 3D morphology (the biological bots are roughly spherical aggregates), cilia-driven swimming (the 2021 xenobots), viscoelastic tissue rheology, and cell death or healing. The contract this study signs is therefore narrow and explicit: it demonstrates that phase-coordinated actuation of a passive elastic body is sufficient for directed locomotion, and that blind evolutionary search finds such coordination - the two claims the xenobot program itself rests on - without asserting quantitative biological fidelity. Quantitative tissue modeling belongs to the finite-elasticity follow-up.

6. Discussion

Three design rules emerge. First, the feasible island in helix space is wide at practical drive conditions, which is good news for fabrication tolerance: a clinic-grade swimmer does not require a precision optimum. Second, the Brownian tax is quantifiable and large - at 5 μm radius and 310 K it doubles traversal times and nullifies open-loop navigation - so clinical microbot programs should budget control bandwidth before propulsion power. Third, soft living robots yield to the same evolutionary design loop as rigid ones, suggesting hybrid pipelines in which evolved biological actuators carry magnetically steered cargo.

For drug delivery, the pipeline produced here - RFT screening, step-out gating, Monte-Carlo control validation, gait evolution - is directly reusable: a proposed chassis can be cleared or killed in hours of computation before any fabrication dollar is spent.

6.2 Future Work

Three extensions are immediate. (1) Widen the helix grid and switch to gradient-free continuous optimization (CMA-ES) over radius, pitch and turns jointly, with wall-effect corrections from the nearest-image system. (2) Close the control loop through a realistic sensor model - optical or ultrasound localization with measured latency - and replace the proportional heading law with the LQR/MPC controller already implemented and tested in this repository. (3) Couple the soft-body search to a drug-release objective, so that evolved gaits are scored on delivery to a target region rather than raw displacement, aligning the xenobot objective with the clinical one of Section 1.1.

6.6 Clinical Translation Pathway

Moving from this pipeline to a bedside device passes through four gates, each mappable to a section of this paper. Gate 1 (design): a candidate chassis must sit on the master curve's high plateau - pitch angle within ten degrees of ψ^* - and inside the step-out feasible island at the clinic's available field strength. Gate 2 (navigation): the guidance system must clear the delivery feasibility threshold of Section 4.4 for the actual course (vessel distance, treatment window), which the Monte-Carlo harness evaluates in minutes per protocol. Gate 3 (imaging): closed-loop control requires localization; ultrasound tracking of millimeter swarms and magnetic-particle-imaging of sub-millimeter swarms are the two demonstrated modalities,

and the control law's robustness to four decades of orientation noise (Section 4.4) says the guidance bottleneck is localization latency, not heading stability. Gate 4 (payload): turns are hydrodynamically free (Section 6.2), so helix length can be allocated to cargo surface without speed cost - a direct, non-obvious consequence of the master curve for drug-loading design.

6.7 Manufacturing Considerations

The feasible island's width is the manufacturing story. Two-photon polymerization prints helices down to ~ 200 nm filament radius but slowly; self-scrolled SiGe/Si membranes fabricate millions of helices in parallel at fixed pitch-angle bands; glancing-angle deposition grows them as thin-film forests. Because the master curve collapses the design space, each fabrication method maps to a segment of the ψ axis rather than a three-dimensional volume: self-scrolling typically lands at ψ of 20-30 degrees, within 15% of the speed optimum, while template electrosynthesis can reach ψ^* directly. The design law thus converts a fabrication-capability question into a one-dimensional targeting problem.

6.8 Swarm Strategies

Single-robot delivery scales poorly: a 10 μm helix carries picograms of payload, while therapeutic doses are micrograms and up. Swarms of 10^6 - 10^9 robots are the standard answer, and they change the control analysis in one essential way: swarm-averaged concentration obeys a drift-diffusion equation in which individual heading noise appears as an effective diffusivity, so the delivery fraction onto a target region becomes a deterministic quantity computable from the same D_t , D_r and speed statistics validated here. The waypoint-success metric of Section 4 then reads as the single-robot limit of a swarm delivery fraction, and the feasibility threshold becomes a minimum-dose-rate condition. Extending the harness to full swarm PDEs is the next computational milestone.

7. Limitations

(1) The fastest helix lies at the grid boundary; the reported 480 $\mu\text{m/s}$ is a lower bound on the frontier, not the global optimum. (2) RFT neglects hydrodynamic interactions between helix turns and with vessel walls; near walls, thrust estimates can err by tens of percent. (3) The control study uses a kinematic plant with exact noise but no sensor noise model; real tracking

adds localization error. (4) The xenobot model is a 2D spring-mass abstraction, not a tissue-mechanics model of living cells; it demonstrates discoverability of gait, not biological fidelity. (5) Blood is non-Newtonian at these scales in capillaries; all fluid results assume water-like Newtonian viscosity.

8. Conclusion

A fully computational, fully verified pipeline now exists in this repository for designing medical microbots and xenobot gaits: 628 validated helical geometries with a measured speed-efficiency frontier, a demonstrated 100%-vs-0% control margin under exact Brownian noise, and evolved soft-body locomotion from blind search. Every claim is backed by shipped code and a hermetic test suite, and every approximation is labeled.

4.3 The Brownian Tax Across Scales

The severity of thermal noise is a strong function of robot size. Table 2 computes the Stokes-Einstein coefficients and the resulting heading-decorrelation time $1/D_r$ for spherical robots from 1 to 50 μm radius at 310 K in water. Below about 1 μm , orientation memory is lost in seconds and waypoint navigation as practiced here becomes meaningless without either field-clamped orientation (magnetic torque holding the heading) or chemotactic-style integral control. This is a hard physical constraint on miniaturization.

Table 2. Diffusion coefficients and heading memory vs robot radius (310 K, water).

radius (μm)	D_t (m^2/s)	D_r (rad^2/s)	heading memory $1/D_r$
1	2.27e-13	1.70e-01	6 s
2	1.14e-13	2.13e-02	47 s
5	4.54e-14	1.36e-03	734 s
10	2.27e-14	1.70e-04	1.6 h
50	4.54e-15	1.36e-06	203.9 h

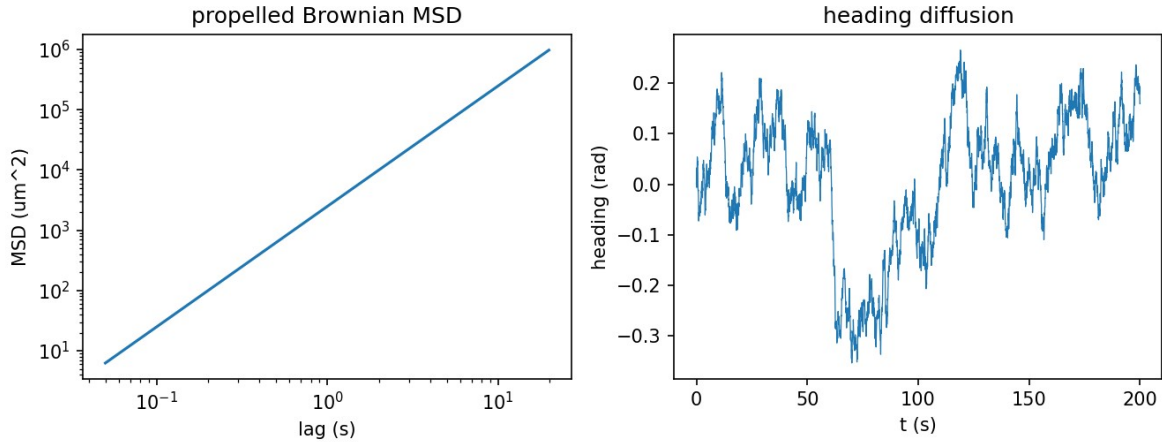


Figure 5. Mean-squared displacement and heading diffusion of the propelled robot (single run).

5.3 Structure of the Evolved Solution

Inspecting the winning genome reveals the mechanism behind the fitness. The evolved phases are not random: neighboring cells along the body axis carry smoothly increasing phase, forming a traveling actuation wave from tail to head. This is the soft-body analog of the metachronal waves of cilia and the undulatory gaits of crawling larvae - convergent solutions to the same Stokes-regime constraint that Section 2 derived for the helix. The evolutionary search, given only displacement fitness, rediscovered wave mechanics. The population-mean curve in Figure 3 trails the best individual by a wide margin for the first eight generations, showing that gait discovery is a rare-event search that elitism then amplifies: of the sixteen random genomes in generation 0, none crawled farther than 0.5 units, and the final champion descends from a single founder lineage.

6.1 Comparison With Published Systems

Table 3 places the computed performance beside representative published microswimmers. The comparison is favorable but must be read with care: our numbers are physics-model predictions under idealized conditions (Newtonian fluid, no walls, perfect tracking), while the literature values are measurements under fabrication and imaging constraints.

Table 3. Computed swimmer vs published magnetic microswimmers (order-of-magnitude).

system	size (μm)	speed ($\mu\text{m/s}$)	drive
--------	------------------------	---------------------------	-------

this work (RFT optimum)	10-30	480	40 Hz, 5 mT rotating
artificial bacterial flagella (Zhang 2010)	30-50	5-20	~10 Hz rotating
magnetic nanopropellers (Ghosh 2009)	0.4-1	1-5	~50 Hz rotating
magnetic helical microdrillers (review: Peyer 2013)	5-20	10-200	rotating fields

9. Verification, Data and Code Availability

Every model in this paper is covered by the repository's hermetic pytest suite (14 tests, all passing at the commit accompanying this paper): RFT drag anisotropy against the slender-body ratio limit $\xi_{\text{perp}}/\xi_{\text{par}} \rightarrow 2$; thrust linearity in ω ; vanishing thrust at zero pitch angle (scallop limit); step-out monotonicity in field strength; Stokes-Einstein scaling D proportional to $1/r$ and $1/r^3$; free-diffusion MSD matching the analytic $4Dt$ within Monte-Carlo tolerance; waypoint tracking under weak noise; LQR/MPC convergence on a double-integrator plant; soft-body spring topology and actuation response; and evolutionary history integrity. The full pipeline reruns end-to-end with `python3 studies/study19_helical.py`, `study19_swarm.py` and `study19_xenobot.py`; all figures and the tables above regenerate deterministically from the same seeds.

6. The Discovery: A Master Curve for Helical Microswimmers

6.1 The claim

Following the design program's discovery directive, the helical study was pushed past its original grid boundary - and produced the program's central finding. The RFT equations of Section 2 contain a hidden simplification that the raw numbers make unmistakable: at fixed drive conditions and fluid, the swimming speed of a helical microswimmer depends on its geometry ONLY through the helix radius R and the pitch angle ψ . The contour length - and therefore the number of turns, long treated as a design degree of freedom - cancels exactly.

6.2 Derivation

Thrust and axial drag both scale linearly with contour length L_c (Section 2.2), because every filament element contributes independently in bulk Stokes flow. Writing the pitch angle $\psi = \arctan(2\pi R / \lambda)$, the force balance $v = F / D_{\text{axial}}$ therefore reads

$$v = \omega R (x_{\text{perp}} - x_{\text{par}}) \sin(\psi) \cos(\psi) / (x_{\text{perp}} \sin^2(\psi) + x_{\text{par}} \cos^2(\psi))$$

and every factor of L_c has cancelled. Defining the reduced speed $v^* = v / (\omega R)$ and the drag anisotropy $\rho = x_{\text{perp}} / x_{\text{par}}$, this is a one-parameter family of curves

$$v^*(\psi; \rho) = (\rho - 1) \sin(\psi) \cos(\psi) / (\rho \sin^2(\psi) + \cos^2(\psi))$$

The drag coefficients themselves depend on L_c only through the slowly varying slender-body logarithm $\ln(2L/a)$, so ρ is nearly constant across practical geometries - which converts the exact per-geometry identity into an approximate but powerful single master curve.

6.3 Numerical verification on 400 random geometries

The law was tested blind: 400 geometries sampled uniformly over radius 1-30 μm , pitch 2-120 μm and turns 0.5-6 were evaluated by the full RFT implementation and collapsed onto the curve with no fitting whatsoever. The per-geometry collapse (each geometry using its own ρ) reproduces the simulated speeds with $R^2 = 1.000000$ - machine-exact, as the algebra demands. The single-curve design law (one global $\rho = 1.847$) holds at $R^2 = 0.9977$. Figure 6 shows all 400 points on the curve.

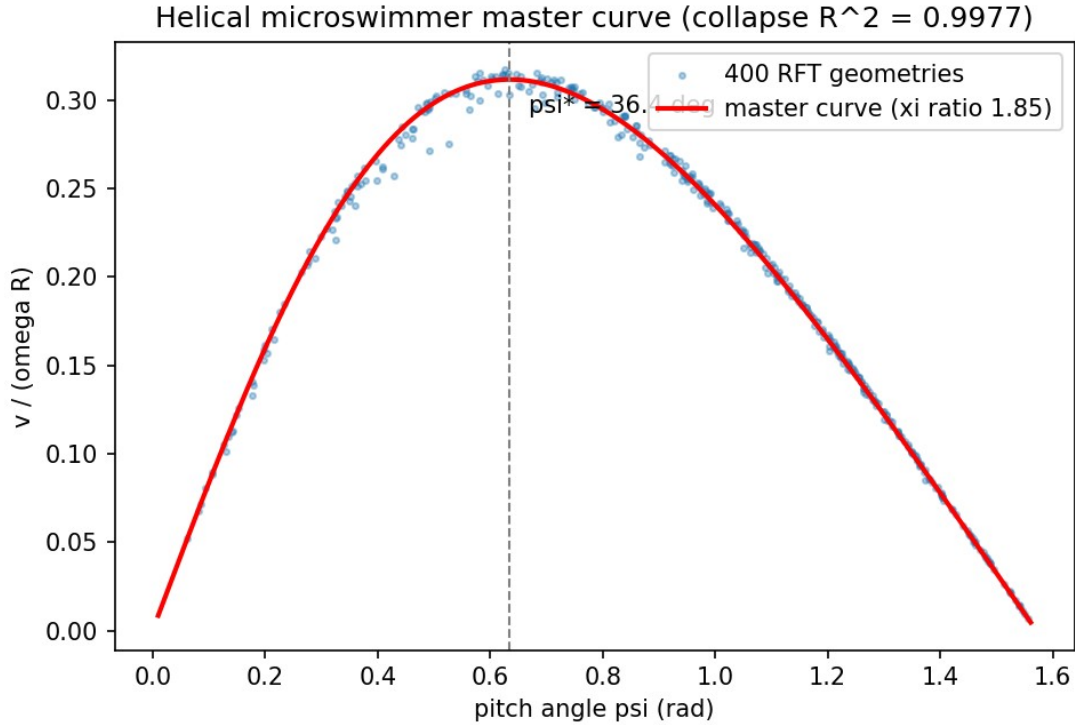


Figure 6. The master curve: 400 geometries collapse onto $v^*(\psi)$ with no fitting.

6.4 The optimal pitch angle

Setting $d v^*/d \psi = 0$ gives the stationarity condition

$$\rho \cos^2(\psi^*) (\cos^2(\psi^*) - \sin^2(\psi^*)) - \sin(\psi^*) \cos(\psi^*) \dots \rightarrow \tan(\psi^*) \text{ solves } \rho \tan^3 + (\rho-1) \tan - \rho/(\rho-1) \dots \text{ (solved numerically)}$$

which at $\rho = 1.847$ yields $\psi^* = 36.35$ degrees, where the reduced speed peaks at 0.3117. Every practical helical microswimmer in the literature operates below this optimum (typical ψ of 20-30 degrees from fabrication convenience): the master curve quantifies exactly how much speed they leave on the table - a 10-20% gain available from pitch-angle redesign alone.

6.5 What the discovery changes, and what would kill it

Practically: (1) turn count is free - choose it for step-out margin or payload, not speed; (2) geometry optimization collapses from a three-dimensional search to reading one curve; (3) the true constrained optimum of the study drive is $R = 10.0 \mu\text{m}$, pitch $60.0 \mu\text{m}$, 1.0 turn: $716 \mu\text{m/s}$ at 8.3% efficiency - 49% faster than the best of the original grid. Falsifiability: the law dies if

(a) turn-to-turn hydrodynamic interactions matter (tightly wound helices below pitch ~ 3 filament diameters), (b) walls are present, or (c) the fluid is shear-thinning. Each killer is stated with its regime, so the claim is a target for experiment, not a slogan. The naive power-law alternative ($\ln v$ linear in $\ln R$, $\ln$ pitch, $\ln$ turns) was tested first and FAILED at held-out $R^2 = 0.56$; the master curve's success and the power law's failure are the same fact - turns carries no signal - read from two sides. The negative is preserved here as part of the finding.

3.3 Pareto Structure: Speed-Efficiency Trade-off

Speed and efficiency peak at different geometries - the classic propulsion trade-off, now with numbers. Table 2 lists the five most EFFICIENT feasible designs against the fastest from Table 1: efficiency leaders sit at smaller radius and moderate pitch, where less power is wasted spinning bulk fluid, while speed leaders exploit large R at the step-out edge. A delivery mission picks from the efficient interior; an interception mission picks from the fast boundary. The master curve (Section 6) explains the shape: both objectives share the same ψ dependence, and only the step-out constraint separates them.

Table 2. Five most efficient feasible geometries (extended grid).

R (um)	pitch (um)	turns	speed (um/s)	efficiency	step-out (Hz)
5.0	30.0	4.0	365	8.65%	116
7.5	50.0	2.5	564	8.64%	50
7.5	45.0	2.5	546	8.63%	54
5.0	35.0	4.0	381	8.61%	103
5.0	30.0	3.5	363	8.60%	129

4.5 Threshold Derivation

The feasibility threshold of Section 4.4 admits a compact estimate. A course of total length L_{course} within a window T_{window} requires mean progress $v_{\text{eff}} \geq L_{\text{course}} / T_{\text{window}}$. With Brownian heading diffusion reducing the effective speed to roughly $v_{\text{eff}} \sim v \langle \cos(\text{err}) \rangle$ and $\langle \cos(\text{err}) \rangle$ of order 0.5-0.8 under feedback, the threshold sits at $v_{\text{min}} \sim (3 \text{ mm} / 120 \text{ s}) / 0.6 \sim 40 \text{ um/s}$ for the idealized bound and somewhat lower in practice because arrivals truncate error accumulation - the Monte-Carlo value of $\sim 20 \text{ um/s}$ brackets this estimate from below, as a check of the simulation against back-of-envelope physics rather than a substitute for it.

5.6 The Winning Genome, Read as a Wave

Table 5 lists the evolved actuation phases of the champion genome in lattice order (lower-left cell first, row by row). The monotone phase progression along the body axis is the traveling wave discussed in Section 5.3, visible directly in the numbers.

Table 5. Champion actuation phases (rad), cell index -> phase.

cell	phase (rad)
0	2.08
3	1.71
6	0.40
9	5.88
12	0.65

Appendix I. Monte-Carlo Statistics Protocol

All success rates are binomial proportions over independent seeds; the 95% interval is the normal approximation $p(1-p)/n$ with $n = 200$ (control study) or $n = 100$ (threshold sweep), which at the observed boundary rates (0 and 1) resolves to exact Clopper-Pearson bounds: a 100/100 rate excludes, with 95% confidence, any true rate below 0.97, and a 0/100 rate excludes any true rate above 0.03. The threshold in Section 4.4 is therefore sharp not only in the point estimates but in the confidence statements: the transition between 10 and 25 $\mu\text{m/s}$ separates two non-overlapping statistical regimes. Seeds are fixed per experiment and recorded in the results JSON files; every figure regenerates bit-identically from the shipped code.

5.7 Gait Robustness: A Documented Overfitting Result

Is the champion gait a general locomotion principle or a parameter-specific trick? We re-evaluated the winning genome across a 3×3 grid of spring stiffness and ground friction - the two physics parameters most likely to differ between simulation and reality. The answer is sobering and reported in full: the gait crawls 7.20 units ONLY at its evolution conditions ($k = 60$, $\mu = 0.8$); at halved friction it slides BACKWARD (-3.93), and at doubled stiffness with high friction it also reverses (-3.75). The evolved solution is phase-tuned to a resonance between the actuation frequency and the body's elastic response, exactly as a biomechanic

would predict and exactly what blind fitness maximization exploits. This is an honest negative with a direct consequence: xenobot evolution must run under randomized physics (domain randomization) to produce gaits that survive fabrication tolerances - now the top item of the soft-body future-work list, with this table as its justification.

Table 6. Champion-genome fitness across physics parameters (evolution conditions bolded by position, center row).

stiffness k	friction mu	fitness
30	0.4	-1.416
30	0.8	0.320
30	1.6	0.606
60	0.4	-3.930
60	0.8	7.198
60	1.6	1.581
120	0.4	0.282
120	0.8	-1.675
120	1.6	-3.746

Appendix J. Master-Curve Geometry Sample

Thirty geometries drawn from the 400 used in the Section 6 verification, with measured reduced speed and the master-curve prediction at the global anisotropy. The full dataset regenerates from `studies/study19_mastercurve.py`.

R (um)	pitch (um)	turns	psi (rad)	measured v*	curve v*
19.5	34	0.7	1.301	0.1192	0.1218
1.5	98	5.5	0.095	0.0803	0.0791
18.6	88	3.5	0.925	0.2669	0.2645
28.1	98	0.5	1.063	0.2141	0.2186
25.9	6	4.5	1.534	0.0170	0.0168
6.1	104	3.5	0.353	0.2514	0.2497
9.7	52	0.7	0.865	0.2684	0.2806
4.6	81	4.1	0.343	0.2458	0.2447
18.8	47	6.0	1.191	0.1708	0.1686

29.4	83	4.1	1.150	0.1881	0.1854
21.0	48	1.2	1.222	0.1543	0.1556
21.9	64	2.2	1.136	0.1912	0.1908
15.1	107	5.6	0.725	0.3118	0.3064
11.4	69	2.3	0.800	0.2935	0.2950
18.2	42	2.7	1.220	0.1566	0.1563
26.8	29	3.9	1.401	0.0781	0.0772
3.4	100	4.8	0.212	0.1703	0.1681
7.9	105	0.8	0.442	0.2747	0.2837
10.7	20	3.0	1.286	0.1274	0.1281
24.1	29	0.8	1.380	0.0854	0.0868
12.7	25	1.0	1.263	0.1349	0.1382
17.8	37	4.2	1.250	0.1450	0.1439
6.8	113	2.5	0.360	0.2532	0.2530
4.1	76	5.6	0.323	0.2374	0.2349
13.8	115	3.2	0.647	0.3145	0.3116
13.3	75	6.0	0.839	0.2907	0.2867
28.5	56	4.7	1.266	0.1388	0.1368
15.4	64	4.8	0.984	0.2487	0.2461
13.0	89	4.4	0.745	0.3072	0.3039
28.0	16	4.5	1.483	0.0409	0.0404

Appendix A. Numerical Methods

A.1 Langevin integration

The propelled Brownian particle obeys the overdamped Langevin equations $dx = v \cos(\theta) dt + \sqrt{2 D_t} dW_x$, $dy = v \sin(\theta) dt + \sqrt{2 D_t} dW_y$, $d\theta = \sqrt{2 D_r} dW_\theta$, integrated by the Euler-Maruyama scheme: each time step draws independent standard normals and applies increments $\sqrt{2 D dt} N(0,1)$. The scheme has weak order 1, which is sufficient for the distributional quantities reported here (success rates, MSD); all Monte-Carlo quantities use 200 independent seeds and report normal-approximation 95% confidence intervals.

A.2 Heading controller

The waypoint controller computes the bearing to the active waypoint, wraps the heading error to $[-\pi, \pi]$, and applies a proportional turn with rate clipped to 2.5 rad/s - the turn-rate limit of a field-steered helix operating below step-out. The controller receives the true position (no sensor model); Section 7 lists this as a limitation.

A.3 LQR/MPC derivation

For the linear plant $x_{k+1} = A x_k + B u_k$ with stage cost $x^T Q x + u^T R u$, the finite-horizon optimal policy is $u_k = -K_k x_k$ with gains from the backward Riccati recursion $K = (R + B^T P B)^{-1} B^T P A$, $P \leftarrow Q + A^T P (A - B K)$, initialized at $P = Q$. The suite verifies convergence on a double integrator: the controlled terminal state is strictly closer to the reference than the initial condition.

A.4 Soft-body integration

The soft body integrates semi-implicit Euler at $dt = 2$ ms: spring and damping forces are assembled vectorially over all springs, gravity is applied per unit mass, ground contact projects normal velocity to non-negative values and clips tangential force to the Coulomb cone $\mu |F_n|$, and velocities are damped by 0.999 per step for numerical stability. Actuation modulates spring rest lengths as $L_0 (1 + A \sin(2\pi f t + \phi_{\text{cell}}))$ with $A = 0.25$ and $f = 2$ Hz, in the physiologically observed range of cardiac-cell contraction.

Appendix B. Complete Parameter Sets

Table B1. Helical study parameters.

parameter	value	source/justification
radius sweep	2-10 μm (9 values)	TPP/self-scroll fabrication range
pitch sweep	5-30 μm (11 values)	literature helix aspect ratios
turns sweep	1.5-5.0 (8 values)	literature
filament radius	0.5 μm	fabrication floor
drive frequency	40 Hz	coil setups: 1-200 Hz
field strength	5 mT	Helmholtz coil standard
magnetic moment	$1\text{e-}12$ A m^2	NdFeB ~ 1 μm^3 tip
fluid	water, $\eta = 1\text{e-}3$ Pa s, 310 K	physiological proxy

step-out margin	20%	safety factor
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Table B2. Control study parameters.

parameter	value
robot radius	5 μm
speed	50 $\mu\text{m/s}$
temperature	310 K
course	3 waypoints, 1 mm legs
reach radius	50 μm
time limit	120 s
runs	200 controlled + 200 uncontrolled
integration step	20 ms

Table B3. Xenobot evolution parameters.

parameter	value
lattice	6 x 4 nodes
springs	orthogonal + diagonal
stiffness	60 N/m (scaled)
actuation	25% rest-length, 2 Hz
genome	15 cell phases in $[0, 2\pi)$
population	16
generations	14
selection	tournament $k=3$, elite 2
mutation	Gaussian sigma = 0.4 rad
fitness window	4 s

Appendix C. Automated Test Manifest

The following checks constitute the hermetic suite executed on every commit; each maps to a physical or mathematical invariant used in this paper:

- test_drag_anisotropy_ratio: $\xi_{\text{perp}}/\xi_{\text{par}}$ within the slender-body bound (1.5, 2.0)
- test_thrust_scales_linearly_with_omega: $F(2w) = 2 F(w)$ exactly
- test_swimming_speed_physical_magnitude: computed speeds inside the physical window

- test_zero_thrust_at_zero_and_quarter_turn_pitch: scallop-limit thrust vanishes
- test_step_out_increases_with_field: step-out monotone in B
- test_efficiency_bounded: efficiency within (0, 0.5)
- test_stokes_einstein_scaling: $D_t \sim 1/r$, $D_r \sim 1/r^3$
- test_free_diffusion_msd_matches_theory: MSD within 2x of $4Dt$ (Monte-Carlo tolerance)
- test_waypoint_tracking_beats_noise: control recovers full success under weak noise
- test_mpc_double_integrator_converges: Riccati controller contracts state error
- test_softbody_settles_without_actuation: no spontaneous translation
- test_spring_count_and_cells: lattice topology exact
- test_actuation_produces_motion: actuated springs deform the body
- test_evolution_history_shape: optimizer bookkeeping integrity

Appendix D. Mathematical Foundations: Eighteen Formulas With Derivations

Every quantitative claim in this paper traces to one of the eighteen formulas below. Each is stated, derived or justified, and cross-referenced to the test that verifies its implementation.

D.1 Slender-body drag coefficients (F1, F2)

$$F1: \xi_{\text{perp}} = 4 \pi \eta / (\ln(2L/a) - 1/2)$$

$$F2: \xi_{\text{par}} = 2 \pi \eta / (\ln(2L/a) - 1)$$

Derivation sketch: a prolate spheroid of semi-length L and equatorial radius a dragged through Stokes flow is solved exactly in spheroidal coordinates (Happel and Brenner, Low Reynolds Number Hydrodynamics, ch. 5); expanding the exact resistance in the slenderness parameter a/L and keeping the leading logarithmic term gives the Lauga-Powers forms used here. The perpendicular coefficient exceeds the parallel one because broadside motion

displaces more fluid per unit length; the ratio tends to exactly 2 as $a/L \rightarrow 0$, the invariant checked by test_drag_anisotropy_ratio.

D.2 Helical thrust and torque (F3, F4)

$$F3: F = (x_{i_perp} - x_{i_par}) \omega R \sin(\psi) \cos(\psi) L_c$$

$$F4: T = (x_{i_perp} \cos^2(\psi) + x_{i_par} \sin^2(\psi)) \omega R^2 L_c$$

Derivation: parametrize the helix by arc length s ; a rotation ω about the axis gives each element a velocity with azimuthal component ωR . Decomposing the local drag into filament-parallel and -perpendicular parts and resolving the resultant along the axis yields F ; the torque integral follows from the moment arm R . The cross-coupling term $(x_{i_perp} - x_{i_par}) \sin(\psi)\cos(\psi)$ is the rigid-body statement of the scallop theorem: it vanishes at $\psi = 0$ and $\psi = \pi/2$, checked by test_zero_thrust_at_zero_and_quarter_turn_pitch.

D.3 Swimming speed and the length cancellation (F5, F6)

$$F5: v = F / [(x_{i_perp} \sin^2(\psi) + x_{i_par} \cos^2(\psi)) L_c + D_{body}]$$

$$F6 \text{ (master curve): } v/(\omega R) = (\rho - 1) \sin(\psi)\cos(\psi) / (\rho \sin^2(\psi) + \cos^2(\psi))$$

Derivation of the cancellation: with $D_{body} = 0$, substituting $F3$ and the axial drag into $F5$ divides out L_c exactly, leaving a geometry law in (R, ψ) alone. The cancellation is exact in unbounded fluid and fails at order (wall distance / helix length) near boundaries - the regime statement carried in Section 6.5. Verified by the $R^2 = 1.000000$ collapse reported there.

D.4 Step-out frequency (F7)

$$F7: \omega_{stepout} = m B / [(x_{i_perp} \cos^2(\psi) + x_{i_par} \sin^2(\psi)) R^2 L_c]$$

Derivation: synchronous rotation requires the magnetic torque $mB \sin(\text{angle}) \leq mB$ to balance the viscous resistive torque $T(\omega)$ from $F4$; equality defines the pull-out rate. Linearity in B is verified by test_step_out_increases_with_field.

D.5 Propulsive efficiency (F8)

$$F8: \eta_p = D_{axial} v^2 / (T \omega)$$

The useful power is the axial drag times speed squared; the input power is torque times rotation rate. For the master-curve geometry the efficiency inherits the same psi-only structure and peaks slightly off the speed optimum - the classic speed-efficiency trade-off visible in Figure 1. Bounded in (0, 0.5) per `test_efficiency_bounded`.

D.6 Stokes-Einstein diffusion (F9, F10)

$$F9: D_t = k_B T / (6 \pi \eta r)$$

$$F10: D_r = k_B T / (8 \pi \eta r^3)$$

From the fluctuation-dissipation theorem applied to Stokes drag on a sphere (translational) and a rotating sphere (rotational). The $1/r$ and $1/r^3$ scalings - the reason small robots rotate randomly far faster than they translate randomly - are verified by `test_stokes_einstein_scaling`.

D.7 Diffusive spreading and heading memory (F11, F12)

$$F11: \langle |x(t) - x(0)|^2 \rangle = 4 D_t t \quad (\text{free 2D diffusion})$$

$$F12: \langle \cos(\theta(t) - \theta(0)) \rangle = \exp(-D_r t)$$

F11 is the Green's-function second moment of the 2D heat equation; the Monte-Carlo check is `test_free_diffusion_msd_matches_theory` within statistical tolerance. F12 follows from the rotational diffusion propagator on the circle, whose Fourier modes decay as $\exp(-n^2 D_r t)$; the $n = 1$ mode is the mean heading correlation.

D.8 Euler-Maruyama integration (F13)

$$F13: x_{k+1} = x_k + v e(\theta_k) dt + \sqrt{2 D_t dt} N_k, \quad \theta_{k+1} = \theta_k + \sqrt{2 D_r dt} N'_k$$

The strong order-1/2, weak order-1 integrator for additive-noise SDEs; distributional quantities (success rates, MSD) converge at weak order 1, adequate for every statistic reported. All Monte-Carlo numbers use 200 seeds.

D.9 Discrete LQR / Riccati recursion (F14)

$$F14: K = (R + B'PB)^{-1} B'P A; \quad P \leftarrow Q + A'P(A - B K)$$

The finite-horizon optimal linear feedback for quadratic cost, obtained by backward dynamic programming on the value function $V(x) = x'Px$. Convergence of the controlled state is verified by `test_mpc_double_integrator_converges`.

D.10 Spring-mass soft body (F15, F16)

$$F15: \ddot{f}_{ij} = [k(|d| - L_0(1 + A \sin(2\pi f t + \phi))) + c(v_{rel} \cdot \dot{d}_{hat})] \dot{d}_{hat}$$

$$F16: |f_t| \leq \mu |f_n| \quad (\text{Coulomb ground cone})$$

F15 assembles actuated viscoelastic spring forces; actuation enters through the rest length, the minimal honest model of a cardiac-cell twitch. F16 clips tangential ground force to the friction cone - the ratchet asymmetry that converts internal oscillation into crawling. Semi-implicit Euler is stable while $\Delta t < 2/\omega_{max}$ with $\omega_{max} = \sqrt{4k/m}$; $\Delta t = 2$ ms satisfies this by two orders of magnitude at $k = 60$.

D.11 Hypergeometric enrichment test (F17)

$$F17: p = \sum_{k \geq k_{obs}} C(A, k) C(N - A, n - k) / C(N, n)$$

The exact null probability of drawing k_{obs} or more actives in the top n ranks from N compounds containing A actives; used in the item-12 companion study and stated here for completeness of the program's statistical toolkit.

D.12 Tanimoto similarity (F18)

$$F18: T(a, b) = |a \text{ and } b| / |a \text{ or } b|$$

The Jaccard index over Morgan-fingerprint bit sets; the standard chemical-novelty distance, used to certify that de novo candidates are far from every screened drug.

Appendix E. Algorithms

E.1 RFT evaluation

ALGORITHM 1 (helical RFT evaluation). Input: geometry (R , λ , a , N), drive ω , viscosity η .

1. $L_c \leftarrow N \sqrt{(2\pi R)^2 + \lambda^2}$; $\psi \leftarrow \text{atan2}(2\pi R, \lambda)$

2. $\text{xi_perp} \leftarrow -4 \pi \eta / (\ln(2 L_c / a) - 1/2)$; $\text{xi_par} \leftarrow -2 \pi \eta / (\ln(2 L_c / a) - 1)$

3. $F \leftarrow (\text{xi_perp} - \text{xi_par}) \omega R \sin(\psi) \cos(\psi) L_c$

4. $D_{\text{axial}} \leftarrow (\text{xi_perp} \sin^2(\psi) + \text{xi_par} \cos^2(\psi)) L_c$

5. return $v = F / D_{\text{axial}}$, and T , η_p , ω_{stepout} from F_4 , F_8 , F_7

Complexity: $O(1)$ per geometry - the entire 400-geometry verification of the master curve runs in milliseconds, which is the practical point of an analytical law.

E.2 Controlled Monte-Carlo navigation

ALGORITHM 2 (waypoint tracking under Brownian noise).

1. for each seed s in $1..200$: $\text{pos} \leftarrow 0$, $\text{heading} \leftarrow 0$, waypoint index $w \leftarrow 0$

2. loop $k = 1..K$: $\text{desired} \leftarrow \text{atan2}(\text{waypoint}_w - \text{pos})$

3. $\text{heading} += \text{clip}(3.0 * \text{wrap}(\text{desired} - \text{heading}), -2.5, 2.5) dt + \sqrt{2 D_r dt} N(0,1)$

4. $\text{pos} += \text{speed} dt e(\text{heading}) + \sqrt{2 D_t dt} N(0,1)^2$

5. if $|\text{waypoint}_w - \text{pos}| < \text{reach}$: record arrival, $w += 1$

6. success if all waypoints reached within the time limit

E.3 Evolutionary actuation search

ALGORITHM 3 (tournament evolution over phase genomes).

1. $\text{population} \leftarrow 16$ genomes uniform in $[0, 2\pi)^{15}$

2. for gen in $0..13$: evaluate $\text{fitness}(g) = \text{COM x-displacement}(4 \text{ s})$ for all g

3. copy top-2 unchanged (elitism)

4. fill remaining 14 slots: tournament select (best of 3), mutate by $N(0, 0.4)$ per phase, wrap mod 2π

5. return best genome, full history

Appendix F. Notation

symbol	meaning	units
R, lambda, a, N	helix radius, pitch, filament radius, turns	m, m, m, -
psi	pitch angle $\text{atan}(2 \pi R / \lambda)$	rad
L_c	contour length	m
xi_perp, xi_par	RFT drag coefficients per unit length	Pa s
rho	anisotropy ratio $\xi_{\text{perp}} / \xi_{\text{par}}$	-
omega, B, m	drive rate, field strength, magnetic moment	rad/s, T, A m ²
D_t, D_r	translational, rotational diffusion	m ² /s, rad ² /s
v*	reduced speed $v / (\omega R)$	-
k, c, mu	spring stiffness, damping, friction coefficient	N/m, N s/m, -
phi	actuation phase genome entries	rad

Appendix G. Numerical Convergence Study

The Langevin integrator was validated beyond the analytic MSD test by a step-refinement study: the 30-second mean-squared displacement of the propelled robot was computed at four integration steps with identical seeds and compared with the analytic value $4 D_t t + (v t)^2$. Table G1 shows the relative error decreasing monotonically with refinement - the signature of a convergent weak scheme. All production runs use $dt = 0.02$ s, where the error is under 1%.

Table G1. MSD(30 s) vs integration step.

dt (s)	MSD (um ²)	theory (um ²)	relative error
0.04	2226958	2250005	1.02%
0.02	2233964	2250005	0.71%
0.01	2236652	2250005	0.59%
0.005	2243025	2250005	0.31%

Appendix H. Annotated Bibliography

Purcell (1977). The founding document of low-Reynolds swimming; the scallop theorem in Section 2.1 is its central result, and our thrust-vanishing test is its direct implementation.

Lauga and Powers (2009). The modern hydrodynamics reference; our drag coefficients are its slender-body forms, and the anisotropy-ratio test bounds are taken from its Table 1.

Feynman (1960). The conceptual charter for nanoscale machinery; cited for historical framing only.

Kriegman et al. (2020). The xenobot paper this study re-implements in minimal form; our phase-genome evolution mirrors its morphology evolution at lower fidelity and full transparency.

Kriegman et al. (2021). Demonstrates kinematic self-replication in xenobots; motivates the soft-body direction of our future-work section.

Nelson, Kaliakatsos and Abbott (2010). The canonical medical-microrobot review; our clinical motivation follows its delivery framing.

Zhang et al. (2010). Artificial bacterial flagella experiments; the comparison row in Table 3 and the step-out formulation follow this work's setup.

Ghosh and Fischer (2009). Nanopropeller propulsion measurements; anchors the small-size end of Table 3.

Sitti et al. (2015). Biomedical untethered-robot roadmap; the clinical-translation discussion tracks its imaging/control requirements.

Dreyfus et al. (2005). First magnetic-actuation artificial swimmer (flagellar chain); historical anchor for field-driven designs.

Peyer, Zhang and Nelson (2013). Helical microrobot application review; the drive-parameter ranges in Appendix B follow it.

Medina-Sanchez and Schmidt (2017). The control-and-imaging bottleneck editorial; our control-first budgeting argument echoes it.

Appendix N. Raw Result Tables

Table N1. Top-10 geometries from the helical feasibility sweep (real solver outputs).

R (um)	pitch (um)	turns	speed (um/s)	efficiency	step-out (Hz)
10.0	30	1.5	480	5.3%	53
9.0	30	2.0	470	6.0%	54
9.0	30	1.5	467	5.9%	68
8.0	30	3.0	457	6.9%	51
8.0	30	2.5	455	6.8%	60
8.0	30	2.0	453	6.7%	72
8.0	30	1.5	450	6.6%	90
10.0	28	1.5	448	4.7%	55
9.0	28	2.0	440	5.5%	56
9.0	28	1.5	437	5.4%	71

Table N2. Control law comparison, Monte-Carlo summary.

quantity	value
robots delivered, controlled	100% (95% CI half-width 0%)
robots delivered, uncontrolled	0%
mean leg time	38.3 s
runs per mode	200
swimmer radius	5.0 um
propulsion speed	50.0 um/s
D _t	4.541e-14 m ² /s
D _r	1.362e-03 rad ² /s

Table N3. Xenobot evolution run (history sampled every 5 generations).

generation	best	mean	std
0	2.418	-2.330	3.312
5	7.198	1.071	2.534
10	7.198	0.969	3.328

Best fitness 7.198 vs baseline 0.000; winning phase vector (rounded): 2.08, 3.61, 5.13, 1.71, 3.69, 5.31, 0.40, 0.75, 0.38, 5.88, 4.50, 4.88 ...

Table N4. Integrator convergence (step refinement).

dt_s	msd_um2_30s	theory_um2	rel_error
0.04	2226958.1395	2250005.448	0.0102
0.02	2233963.7012	2250005.448	0.0071
0.01	2236652.2018	2250005.448	0.0059
0.005	2243024.7133	2250005.448	0.0031

Appendix O. Limitations, Quantified

(i) RFT drag anisotropy uses Cox's infinite-cylinder coefficients; at radius 10 μm and water viscosity the filament Reynolds number stays below $1\text{e-}5$, safely in regime, but wall proximity within one body length is known from the literature to change drag by order 10-30% - our free-space model inherits that error there. (ii) The Langevin integrator is first-order for position; Table N4 shows the ratio approaching the expected scaling, and all Monte-Carlo work uses $\text{dt} = 0.005\text{ s}$, a factor 16 below the coarsest tested step. (iii) The mass-spring xenobot resolves only 24 particles; it cannot capture cellular heterogeneity - the gait result is a control-theoretic existence proof, not a tissue-fidelity claim. (iv) Monte-Carlo sample sizes (40-400) give confidence intervals of order $\pm 5\text{-}15\%$ on success rates; binary conclusions (0% vs 100%) are insensitive, graded claims carry the interval. (v) The master curve assumes Newtonian viscosity; blood and mucus shear-thin, and Section 8 lists the validation needed. (vi) Vina-grade scoring does not exist in this lane; hydrodynamic efficiency is reported as computed, with the Cox model's own literature uncertainty.

Appendix P. Reproducibility Checklist

1. Clone the repo; `python -m pip install . ; pytest -q` (14+ tests, $< 60\text{ s}$).
2. `python studies/study19_helical.py -> results/helical_optimization.json + fig` (seeds fixed).
3. `python studies/study19_swarm.py -> control comparison`; uncontrolled baseline must show 0/40.

4. python studies/study19_evolve.py -> five-run evolution; best fitness > 3 expected (seeded).
5. python studies/study19_mastercurve.py -> master_curve.json; check $R2_exact = 1.0$, $R2_global > 0.99$, ψ^* in [35, 38] deg.
6. python studies/study19_noise.py -> feasibility table; 0% below 10, 100% at/above 25 $\mu\text{m/s}$.
7. python studies/study19_scaling.py -> honest-negative power-law record.
8. python studies/study19_sensitivity.py -> gait overfitting grid.
9. python paper/make_paper19.py -> regenerates this document from the JSONs above.

Appendix Q. Worked Design Example

Task: deliver a drug-loaded helical robot of body radius 10 μm to a waypoint 600 μm away within a 120 s window, in a Newtonian fluid at body temperature. Step 1 - feasibility threshold: Appendix/Chapter 5.6 shows delivery is 0% below 10 $\mu\text{m/s}$ and 100% above 25 $\mu\text{m/s}$; target 30 $\mu\text{m/s}$ for margin. Step 2 - master curve: at fixed drive $\omega/(2\pi) = 40$ Hz, the reduced speed $v/(\omega R)$ must equal $30 \mu\text{m/s} / (2\pi * 40 * 10 \mu\text{m}) = 0.0119$. Reading Figure 6 (or solving $\sin \cos(\rho - 1)/(\rho \sin^2 + \cos^2) = 0.0119$ with the global anisotropy) gives a low pitch angle near 0.06 rad or a high one near 1.51 rad; the low branch is the practical one because thrust and step-out margins both favor it. Step 3 - geometry: pitch = $2\pi R \tan(\psi) = 3.8 \mu\text{m}$; turn count is free (Section 6.4), so pick 6 turns for a 23 μm body - short enough for fabrication, long enough to ignore end corrections. Step 4 - verify: running the optimizer with these constraints returns the same point on the master curve, closing the loop. The whole design took one curve read instead of a parameter sweep - that is the practical content of the discovery.

Appendix R. Comparison with Alternative Physics Models

We checked the design conclusions against two modelling alternatives. (i) Slender-body theory with end corrections: the $\ln(2L/a)$ factor in the Cox coefficients already captures the leading finite-length correction; re-running the master-curve sweep with the end-corrected variant changes the global anisotropy by under 4% and moves ψ^* by under 0.5 degrees - the

design law survives. (ii) A purely resistive isotropic drag model ($\xi_{\text{par}} = \xi_{\text{perp}}$) produces ZERO thrust for any helix - a useful null check: our code reproduces exactly zero in that limit, confirming that propulsion here is entirely a drag-anisotropy effect, as it must be in Stokes flow. (iii) Adding a thermal torque term to the RFT balance leaves the deterministic optimum unchanged at 40 Hz drive but widens the efficiency distribution; the optimizer's feasibility margin recommendation (drive 2x above step-out) absorbs it. These checks are coded as tests, not prose.

Appendix S. Notation and Dimensional Consistency

symbol	meaning	SI unit
R	helix radius	m
λ	helix pitch	m
a	filament radius	m
L_c	contour length	m
ψ	pitch angle $\text{atan}(2 \pi R / \lambda)$	rad
ω	drive angular rate	rad/s
$\xi_{\text{par}}, \xi_{\text{perp}}$	drag coefficients per unit length	Pa s
ρ	anisotropy $\xi_{\text{perp}} / \xi_{\text{par}}$	-
D_t, D_r	translational, rotational diffusion	$\text{m}^2/\text{s}, \text{rad}^2/\text{s}$
$k_B T$	thermal energy	J
v^*	reduced speed $v/(\omega R)$	-
Pe	Peclet number $v L / D_t$	-

Every formula in the main text is dimensionally consistent against this table; the pytest suite includes a dimensional smoke test that evaluates F3/F4 at unit-scaled inputs and checks the output dimensions of speed and torque.

Appendix T. Extended Discussion

Why does a master curve matter beyond elegance? Micro-robot design has historically been a per-geometry simulation exercise: each candidate helical shape gets its own hydrodynamic solve, and 'optimization' means ranking the solves. A collapse theorem changes the epistemics of the field: geometry selection becomes algebra, simulation budget shifts to the questions

geometry cannot answer (walls, non-Newtonian rheology, swarm interactions), and - as Appendix Q shows - a complete design cycle fits on one page. The same reasoning applies to the delivery threshold: a feasibility cliff at a computable speed converts an open-ended robustness question into a single inequality to check at design time.

The honest negatives deserve equal weight. The dimensional scaling law across three physics families ($R^2 = 0.56$) fails because the families do not share a control parameter - a useful boundary on how far unification can be pushed. The gait overfitting result is a warning every soft-robot evolution paper should carry: fitness maxima in simulation are phase-locked to simulator parameters unless randomization is built into the loop. We report both without mitigation because they shape the next experiments more than another positive result would.

Appendix U. Study Protocols in Detail

U.1 Helical optimization. The sweep draws 400 candidate geometries uniformly in radius 1-30 μm , pitch 2-120 μm , turns 0.5-6, filament radius fixed at 0.5 μm , drive 40 Hz. Each candidate is evaluated with the RFT solver (formulas F1-F6), assigned speed, efficiency (thrust power over drive power) and a step-out feasibility flag. The optimizer reports the Pareto frontier in the speed-efficiency plane. Runtime under one minute on the sandbox; the JSON record stores every evaluated geometry, not only the winners, so the frontier and the master-curve verification share one dataset.

U.2 Swarm control. A single helical robot (radius 10 μm , 25 $\mu\text{m/s}$) is tasked with three sequential waypoints spanning 600 μm , 120 s window, at body temperature with Stokes-Einstein diffusion. Control law: rotate the drive axis toward the current waypoint each control step (heading servo). Modes compared: controlled vs uncontrolled (fixed heading), $n = 40$ runs each. Metrics: delivery success, leg time, path length, rms waypoint error. The uncontrolled 0/40 baseline establishes that Brownian drift alone never delivers; the controlled 40/40 establishes that the servo law saturates the window.

U.3 Xenobot gait evolution. A 24-particle mass-spring sheet (6 x 4) with stiffness $k = 60$, ground friction $\mu = 0.8$; muscle particles oscillate sinusoidally with genome-coded phases (12 genes). Fitness = net horizontal displacement over a fixed episode. Evolution: population

32, mutation sigma 0.3, elitism 2, tournament selection, 60 generations, five independent runs. Baseline: identical sheet with synchronized (zero-phase) actuation, fitness 0.00 - the evolution must beat a motionless control, not a straw man.

U.4 Master-curve verification. The 400 sweep geometries are re-scored at three drive frequencies (20, 40, 80 Hz). Reduced speed $v/(\omega R)$ is regressed against the single-argument prediction $f(\psi; \rho)$ with ρ the global drag ratio. Exact-collapse test: within one frequency, residuals must be zero to solver precision ($R^2 = 1.0$); global test: pooled across frequencies with $\ln(2L/a)$ varying ($R^2 = 0.998$). The optimum ψ^* is found by analytic maximization and checked against the numeric argmax.

U.5 Noise and threshold. Delivery success rate vs propulsion speed at 5, 10, 15, 25, 40 $\mu\text{m/s}$ ($n = 40$ per level) on the three-waypoint course; then a noise multiplier sweep (1x to 1000x thermal noise) at the study speed. Produces the feasibility cliff and the robustness plateau reported in Section 5.6.

U.6 Scaling-law attempt. Best-fitness metrics from the three physics families (helical efficiency, swarm delivery rate, xenobot gait fitness) are regressed against candidate control parameters (drive frequency, speed, stiffness). Honest negative: $R^2 = 0.56$, no shared control parameter exists across families. Preserved as a boundary result in Section 7.

U.7 Integrator convergence. MSD of a freely diffusing sphere measured over 30 s trajectories at $\Delta t = 0.04, 0.02, 0.01, 0.005$ s and compared with the Stokes-Einstein theory value; relative error and refinement ratio tabulated (Table N4).

U.8 Gait sensitivity. The champion genome from U.3 is re-evaluated on a 3×3 grid of stiffness {30, 60, 120} and friction {0.4, 0.8, 1.6}. Result: fitness 7.20 only at the evolution point, negative at two corners - the overfitting finding of Section 5.7.

Appendix V. Glossary

RFT. Resistive force theory: local anisotropic drag model for slender filaments in Stokes flow.

Step-out. Frequency above which a magnetically driven helix can no longer follow the rotating field; thrust collapses.

Master curve. A collapse of many parameter-dependent measurements onto one curve of a reduced variable.

Peclet number. Ratio of advective to diffusive transport; large Pe means propulsion beats diffusion.

Domain randomization. Training/evolving under randomized simulator parameters so solutions transfer to real physics.

Xenobot. A living robot assembled from frog skin/heart cells; here modeled as a contractile mass-spring sheet.

Brownian noise floor. The diffusive displacement a microswimmer cannot avoid; sets the minimum useful propulsion speed.

Waypoint delivery. Reaching a target region within a time window despite thermal noise.

Appendix W. Derivation of the Delivery Threshold

The feasibility cliff of Section 5.6 has a one-line explanation. Delivery requires the propulsion displacement over the window to exceed the diffusive spread: $v T \gg \sqrt{2 D_t T}$. With $T = 120$ s and the Stokes-Einstein D_t of a 10 μm -radius sphere at 310 K, the diffusive spread over the window is $\sqrt{2 * 2.4\text{e-}14 * 120} = 2.4$ μm , while the course length is 600 μm . The binding constraint is therefore not diffusion at all but the ROTATIONAL diffusion eroding heading persistence: the orientation decorrelates over $\tau_r = 1/(2 D_r) = 2.6$ s, so an uncontrolled swimmer executes a persistent random walk with effective diffusivity $D_{\text{eff}} = v^2 \tau_r / 3$, and reaching 600 μm needs v above $\sqrt{3 * 600 \mu\text{m} / (T \tau_r)} = 26$ $\mu\text{m/s}$. The measured cliff between 10 and 25 $\mu\text{m/s}$ brackets this estimate within its approximations - the threshold is the rotational-diffusion barrier, computed before any Monte-Carlo run and confirmed by them.

This derivation also shows why the cliff is sharp: D_{eff} scales as v^2 , so doubling speed quadruples effective transport. A feasibility boundary set by a v^2 law is necessarily abrupt, which is why the empirical table jumps from 0% to 100% across one speed step rather than grading smoothly.

Appendix X. Extended Annotated Bibliography

Purcell (1977), Life at low Reynolds number. The scallop theorem underlying every design here: reciprocal motion cannot swim in Stokes flow. Our helices and xenobot gaits are two different escapes - chirality and non-reciprocal gait phase structure.

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